

ANALYZER

Logic Analyzer for RCBUS using PI PICO 2[W]

PAGE 1 : INTRODUCTION

RcBusInterface

File: RcBusInterface.kicad_sch
PiPicoA

File: PiPcioA.kicad_sch
PiPicoB

File: PiPcioB.kicad_sch
PiPicoC

File: PiPcioC.kicad_sch

PAGE 2 : RCBUS INTERFACE & POWER SUPPLY

PAGE 3 : PI PICO ANALYZER A

PAGE 4 : PI PICO ANALYZER B

PAGE 5 : PI PICO ANALYZER C

BACKGROUND OF DESIGN

This design is a derivative work of John Winans S-100 bus logic analyzer targeted for RCBUS instead of S-100 bus.
<https://github.com/johnwinans/2070-S-100-bus-analyzer>

Both projects use the Dr Gusman software and firmware.
<https://github.com/gusmanb/logicanalyzer>

PCB STACKUP NOTE

JLC04161H-3313 stackup gives :

- Ideal trace impedance (50 & 100ohms).
- GND plane closer to signal layer routing for improved signal integrity.

MOUNTING/TOOLING HOLES

LOG01

LOG02



H4
MountingHole

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Logic Analyzer for RCBUS

Title: Logic Analyzer

Size: USLetter

Date: 2026-04-27

Rev: 0

KiCad E.D.A. 10.0.1

Drawn: Denno Wiggle

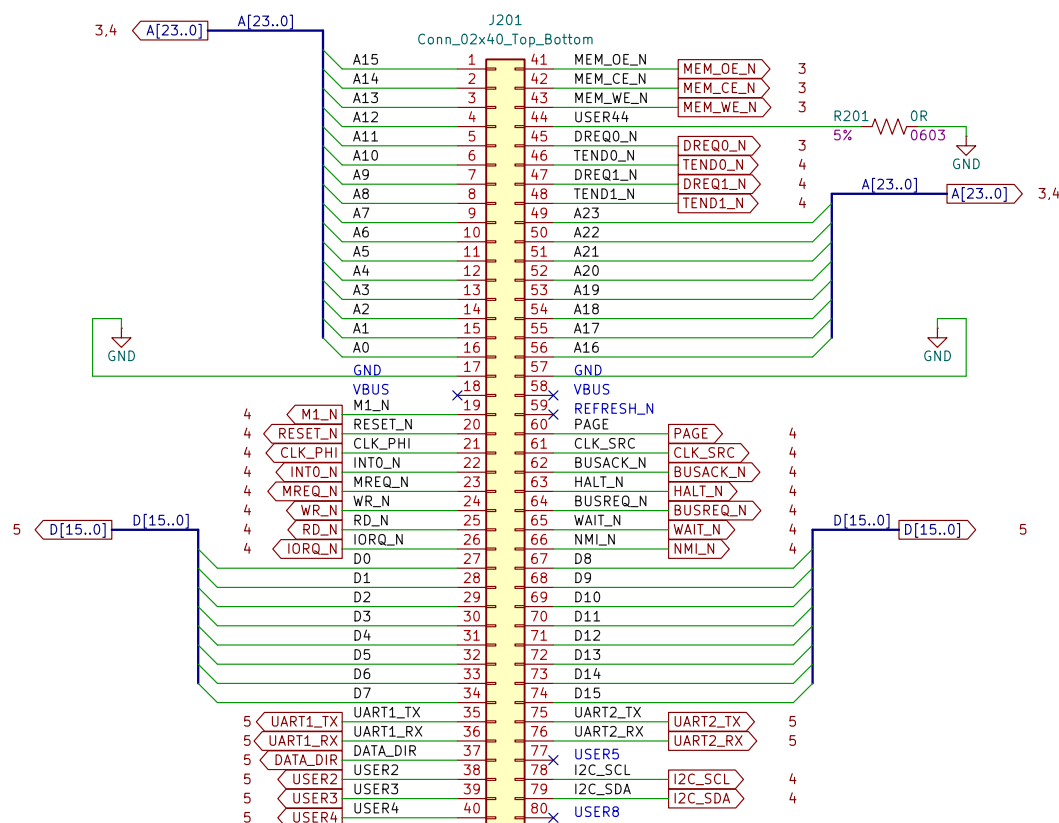
Id: 1/5

RCBUS CONNECTOR

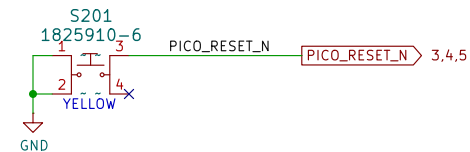
RCBUS CARD SOCKET CONNECTOR

40-PIN LEFT SIDE

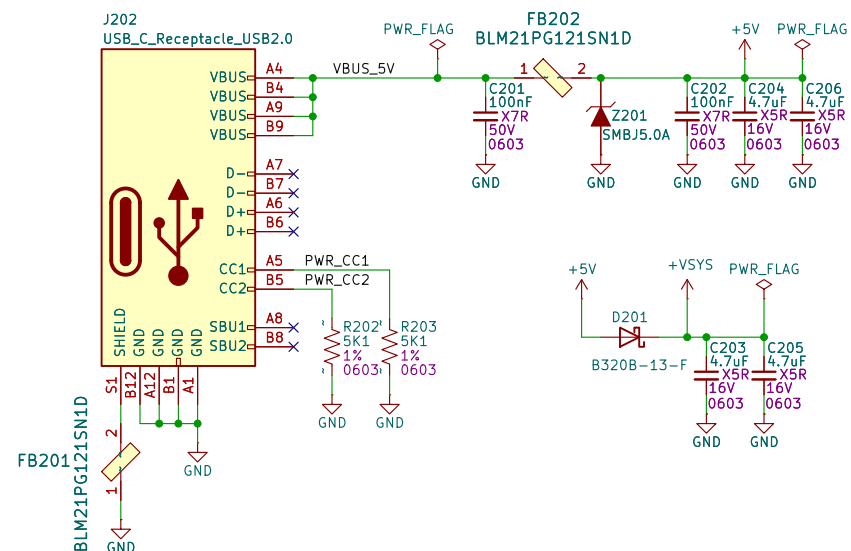
80-PIN BOTH SIDES



PUSH BUTTON RESET FOR PICO



5.0V USB POWER SUPPLY

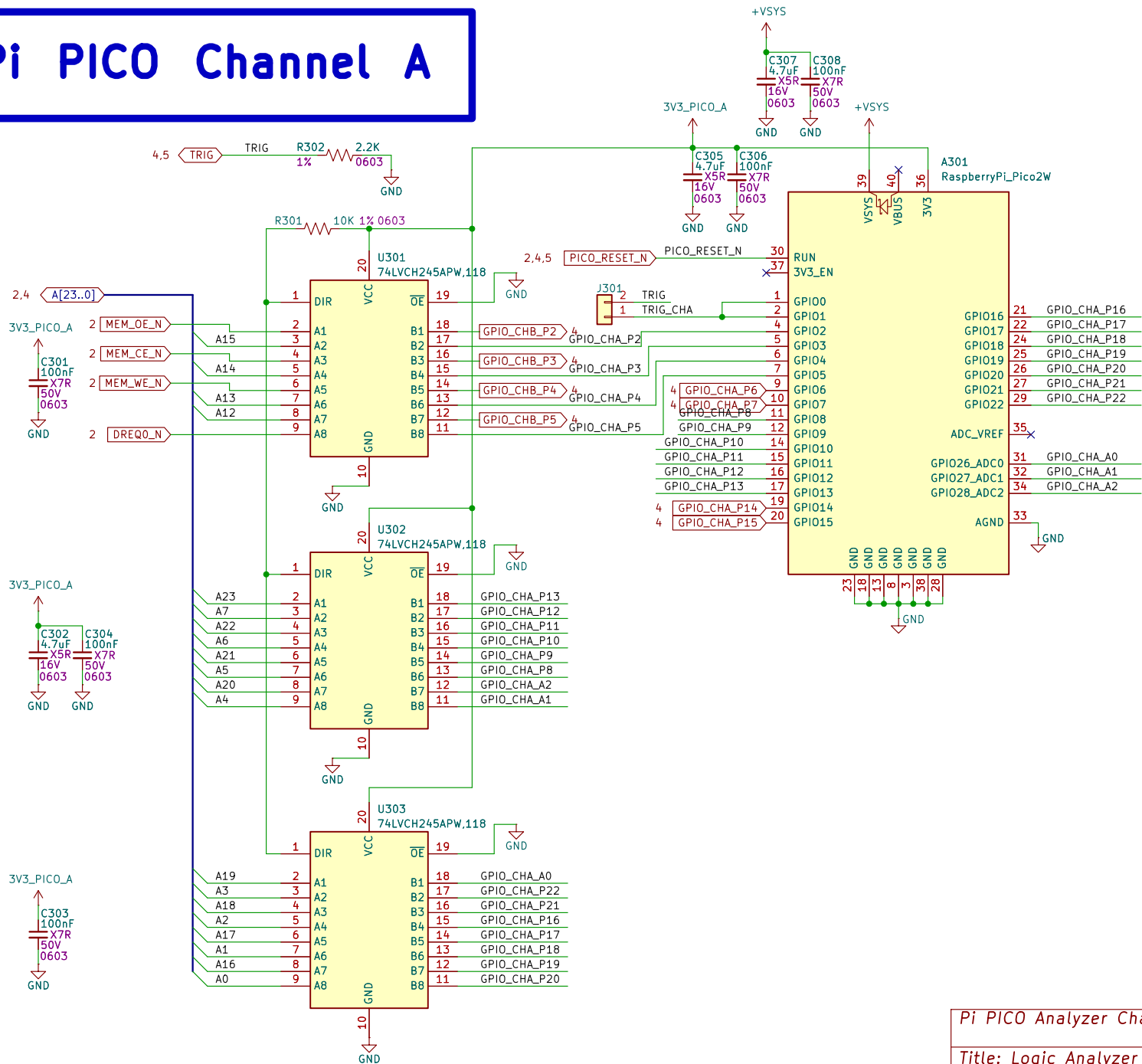


RC Bus Connections and Power Supply

Title: Logic Analyzer

Size: A	Date: 2026-04-27	Rev: 0
KiCad E.D.A. 10.0.1	Drawn: Denno Wiggle	Id: 2/5

Pi PICO Channel A

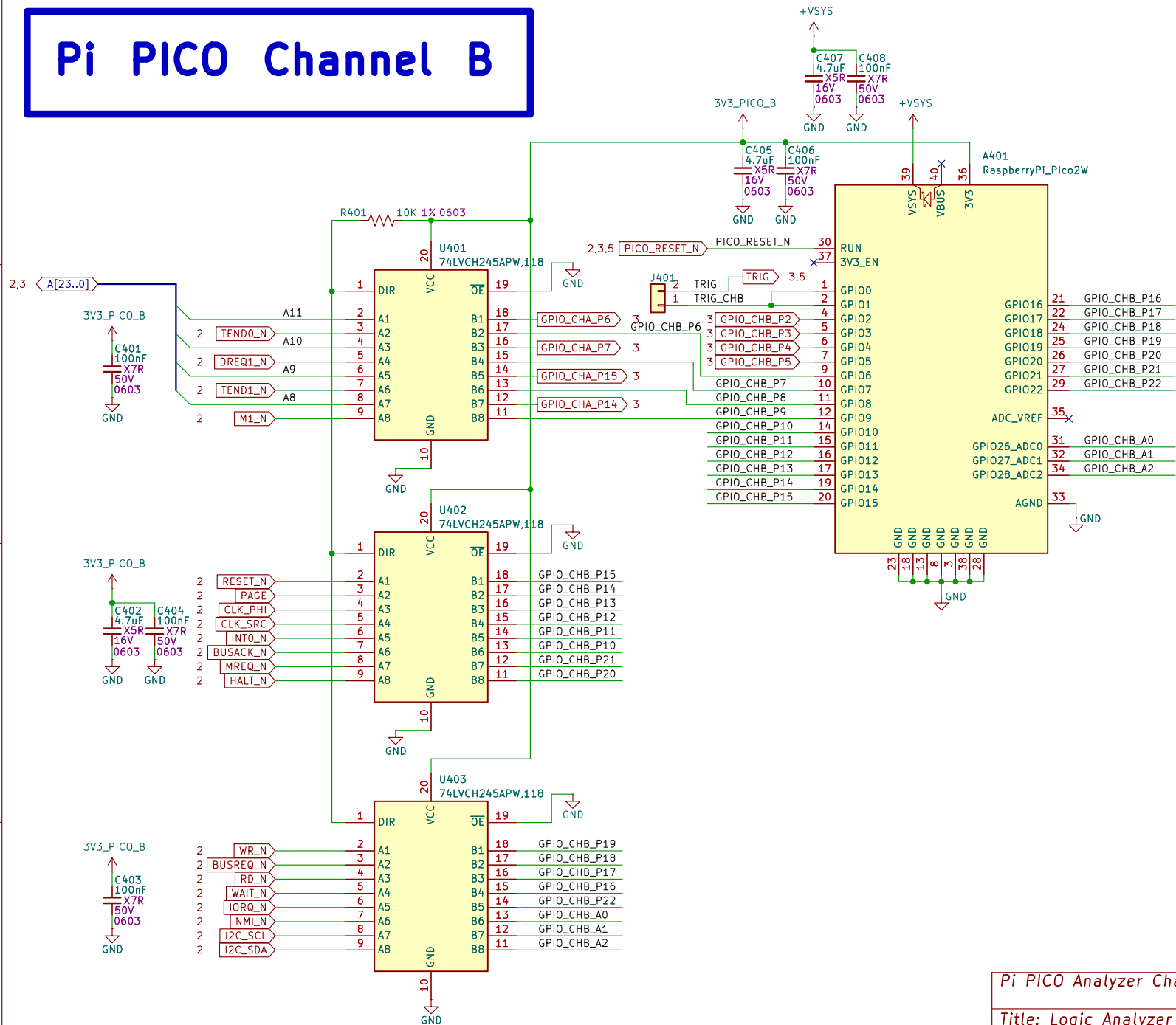


Pi PICO Analyzer Channel A

Title: Logic Analyzer

Size: USLetter	Date: 2026-04-27	Rev: 0
KiCad E.D.A. 10.0.1	Drawn: Denno Wiggle	Id: 3/5

Pi PICO Channel B



Pi PICO Analyzer Channel B

Title: Logic Analyzer

Size: USLetter

Date: 2026-04-27

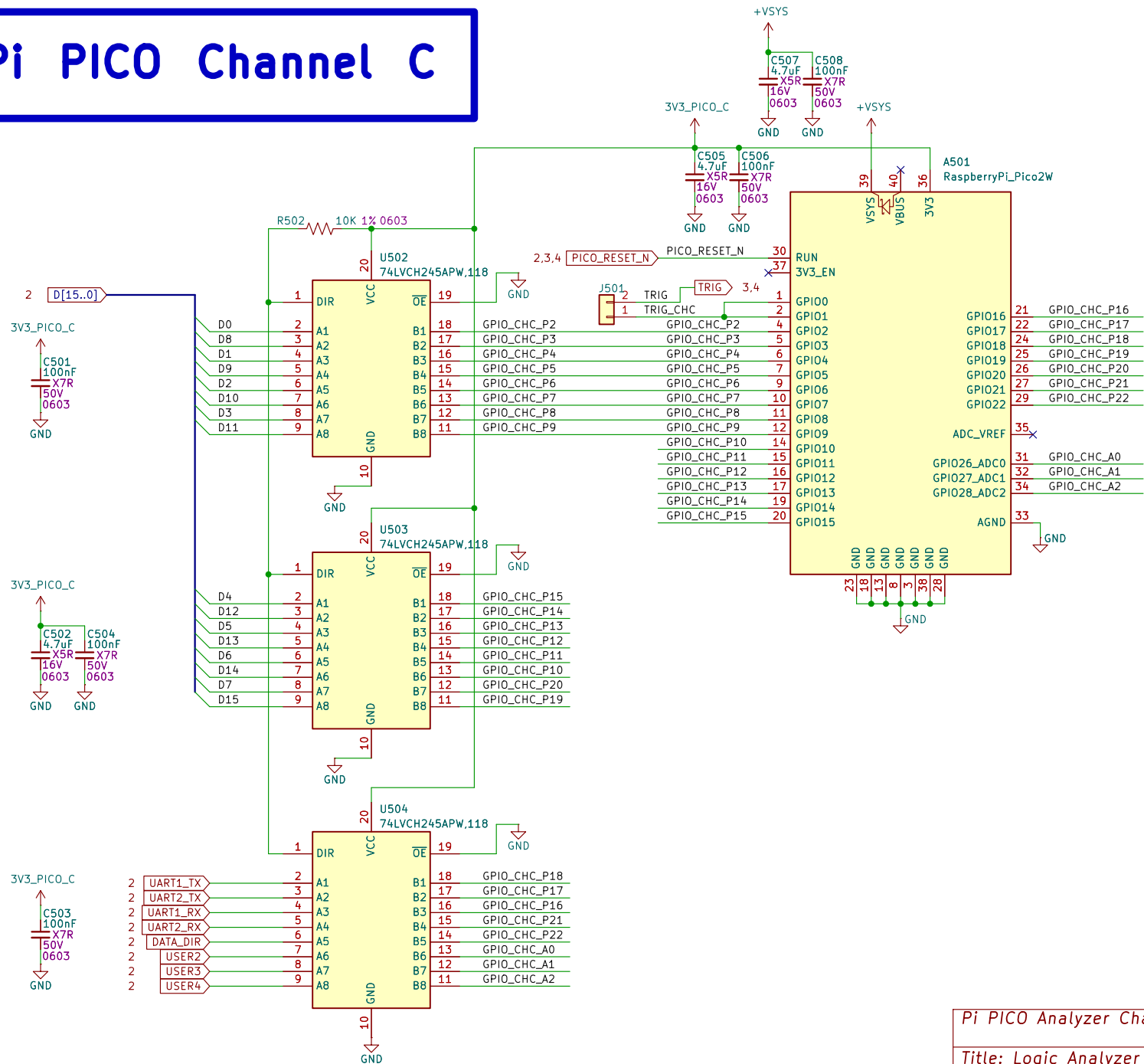
Rev: 0

KiCad E.D.A. 10.0.1

Drawn: Denno Wiggle

Id: 4/5

Pi PICO Channel C



Pi PICO Analyzer Channel C

Title: Logic Analyzer

Size: USLetter

Date: 2026-04-27

Rev: 0

KiCad E.D.A. 10.0.1

Drawn: Denno Wiggle

Id: 5/5